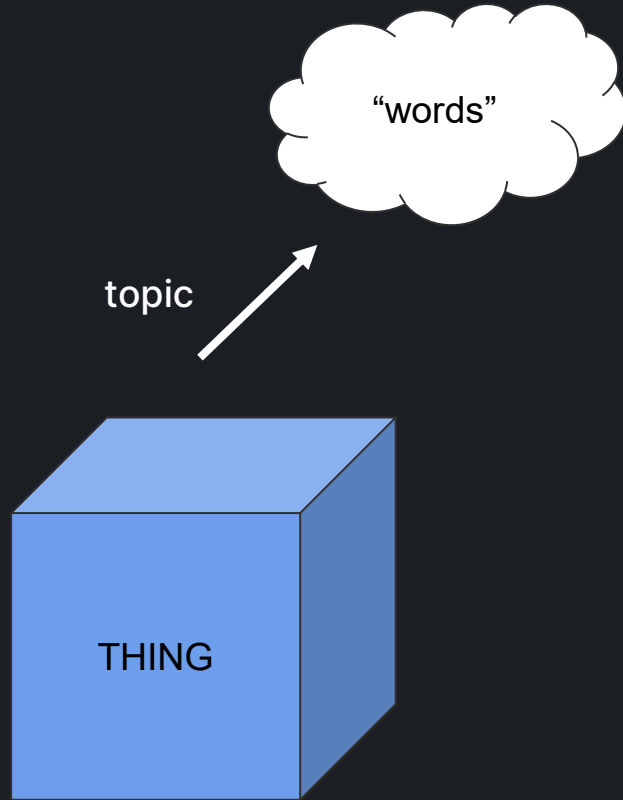




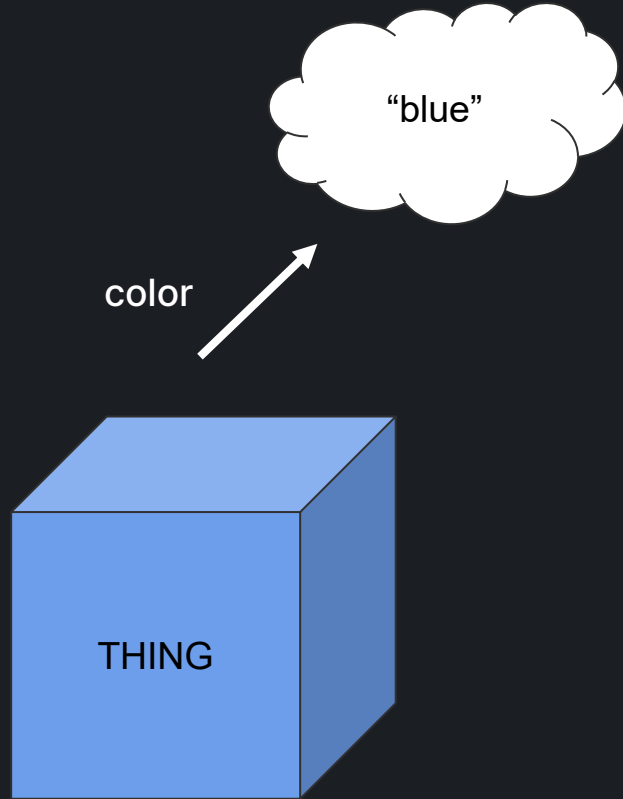
Extract-Edit-Apply



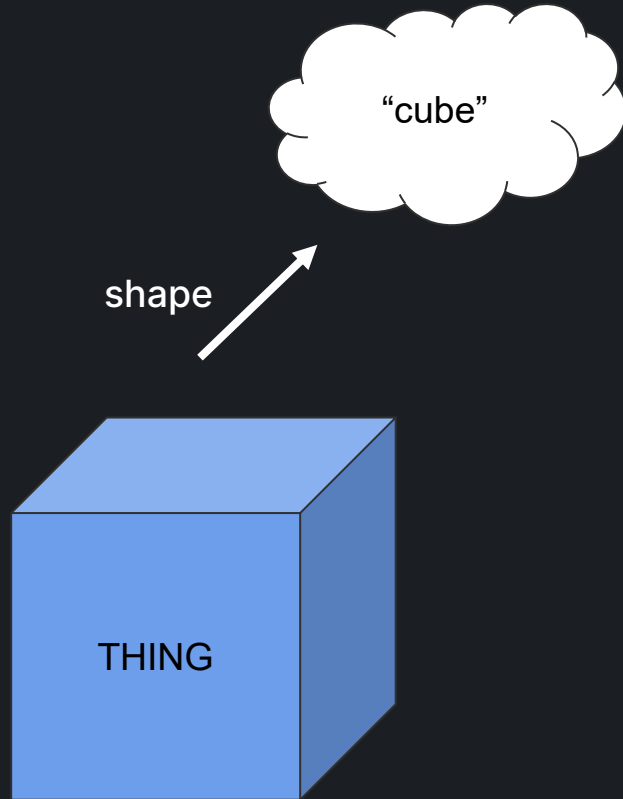
Philosophy



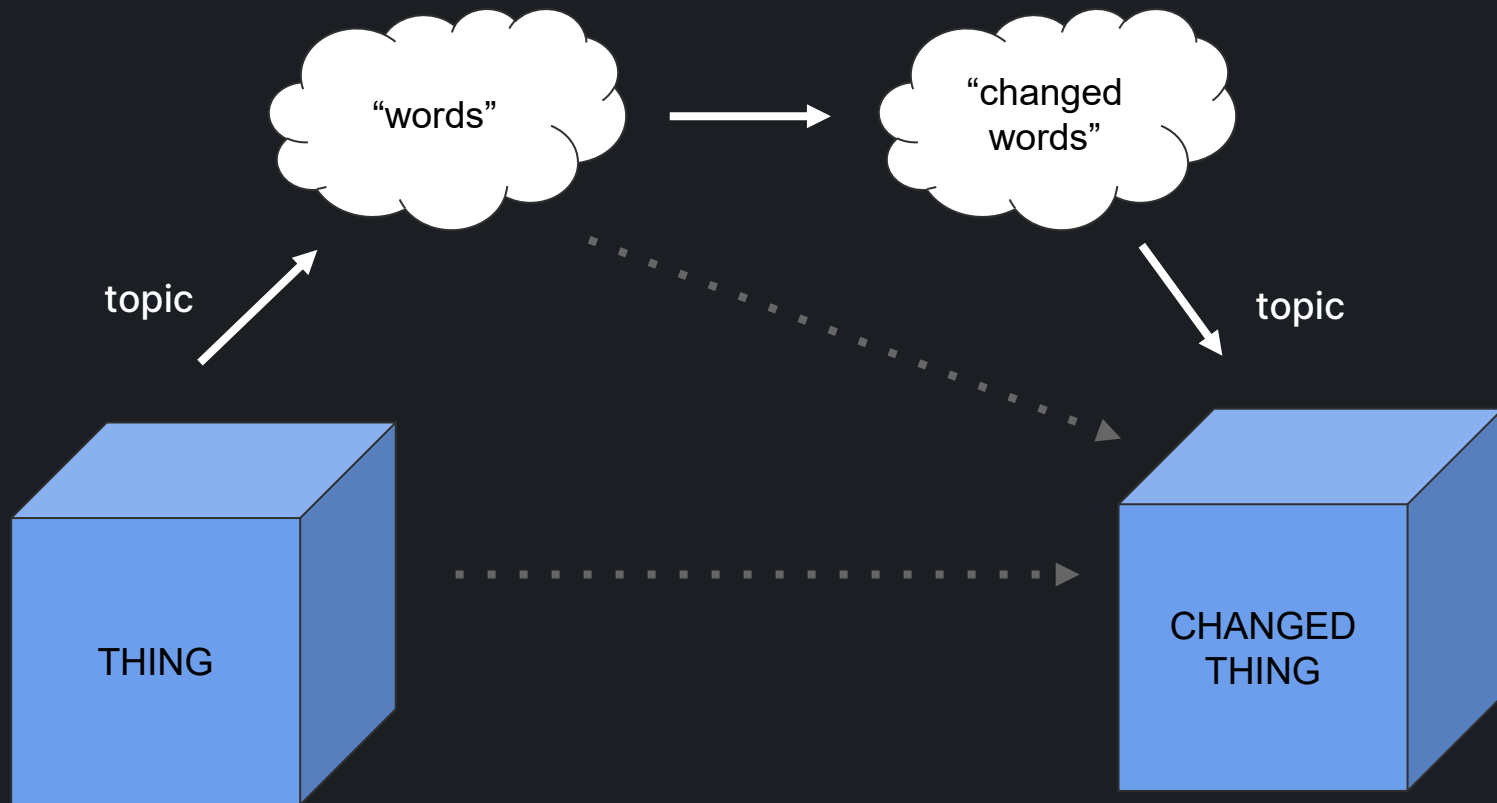
Philosophy



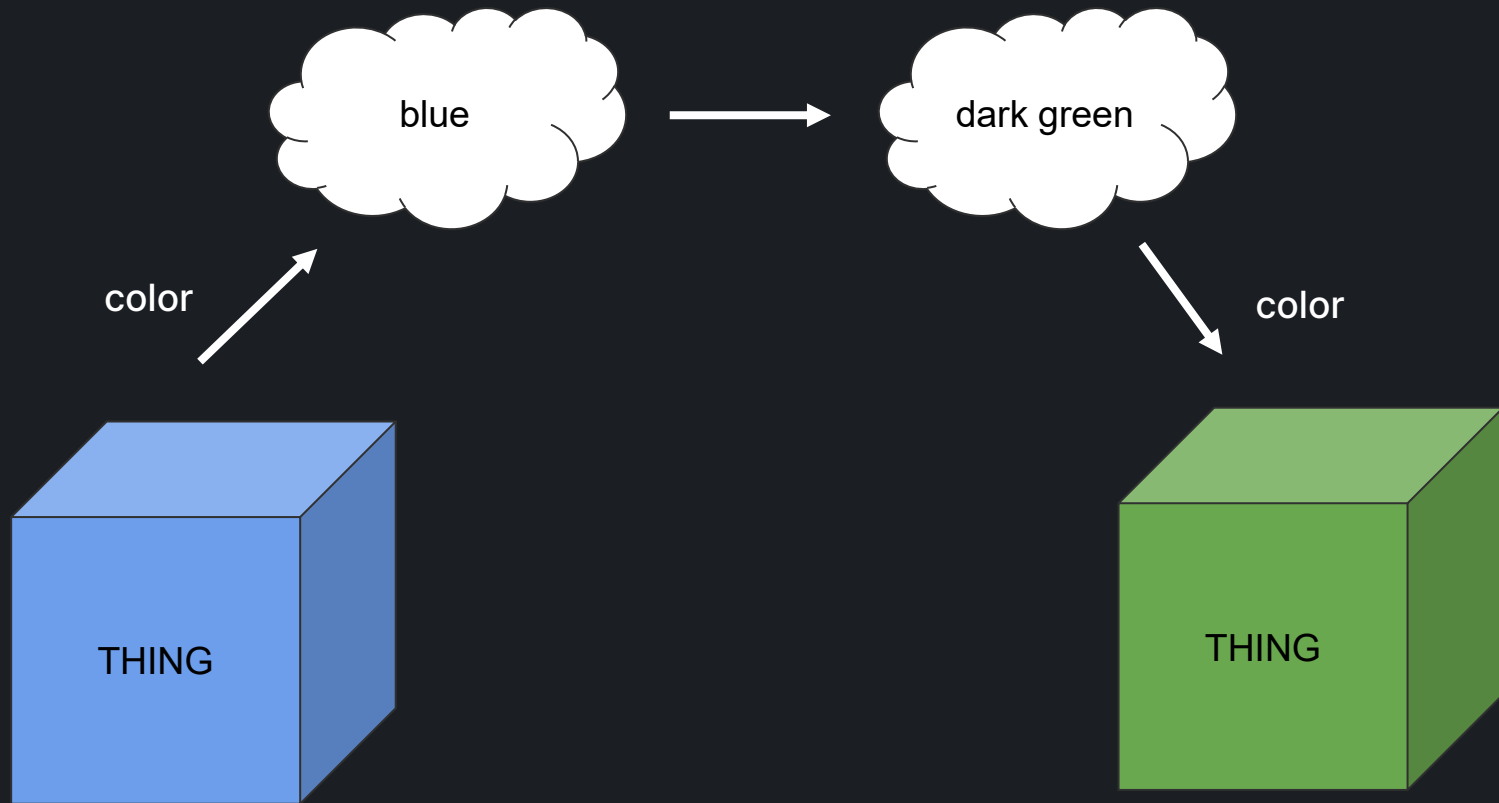
Philosophy



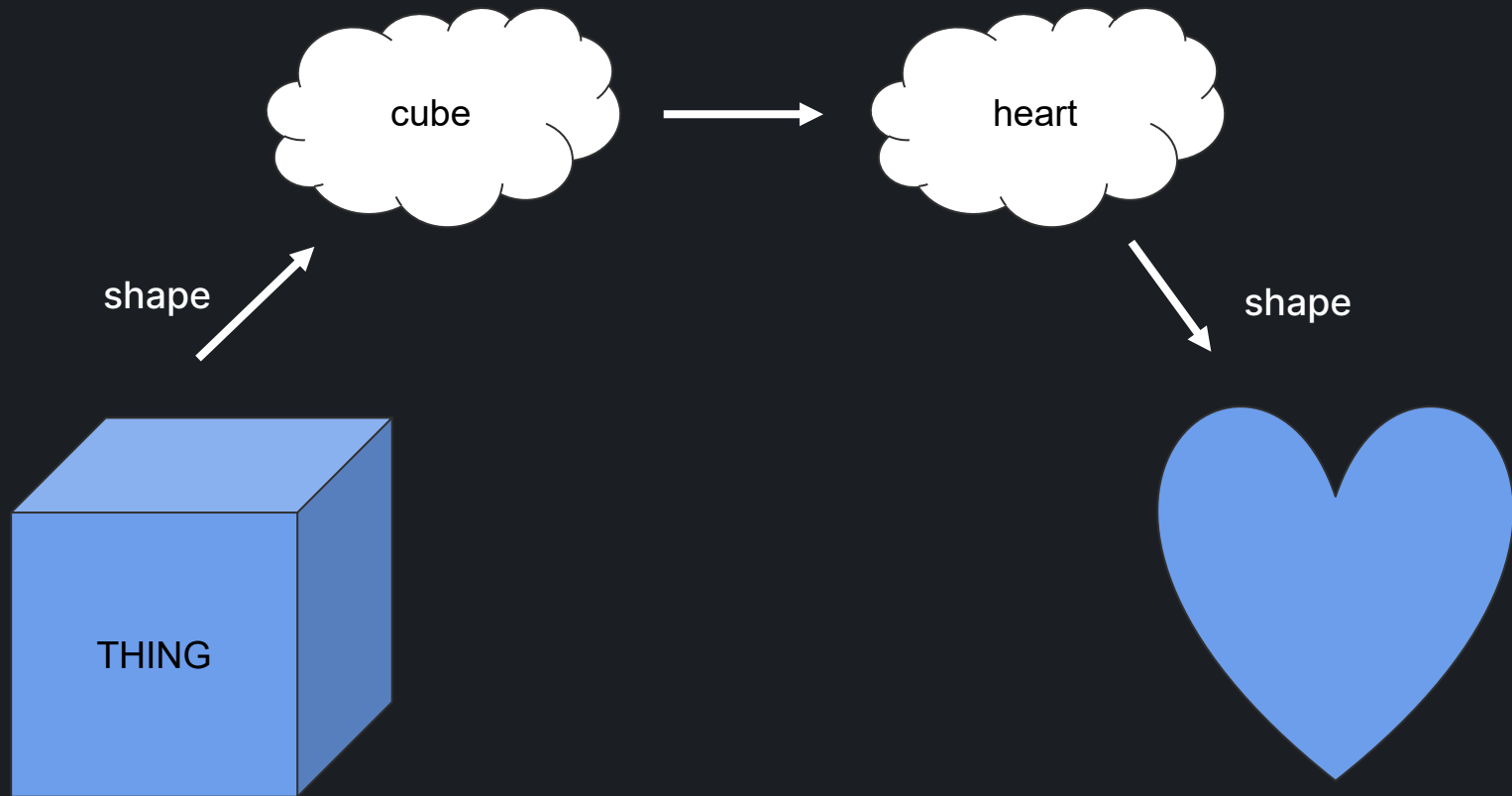
Philosophy



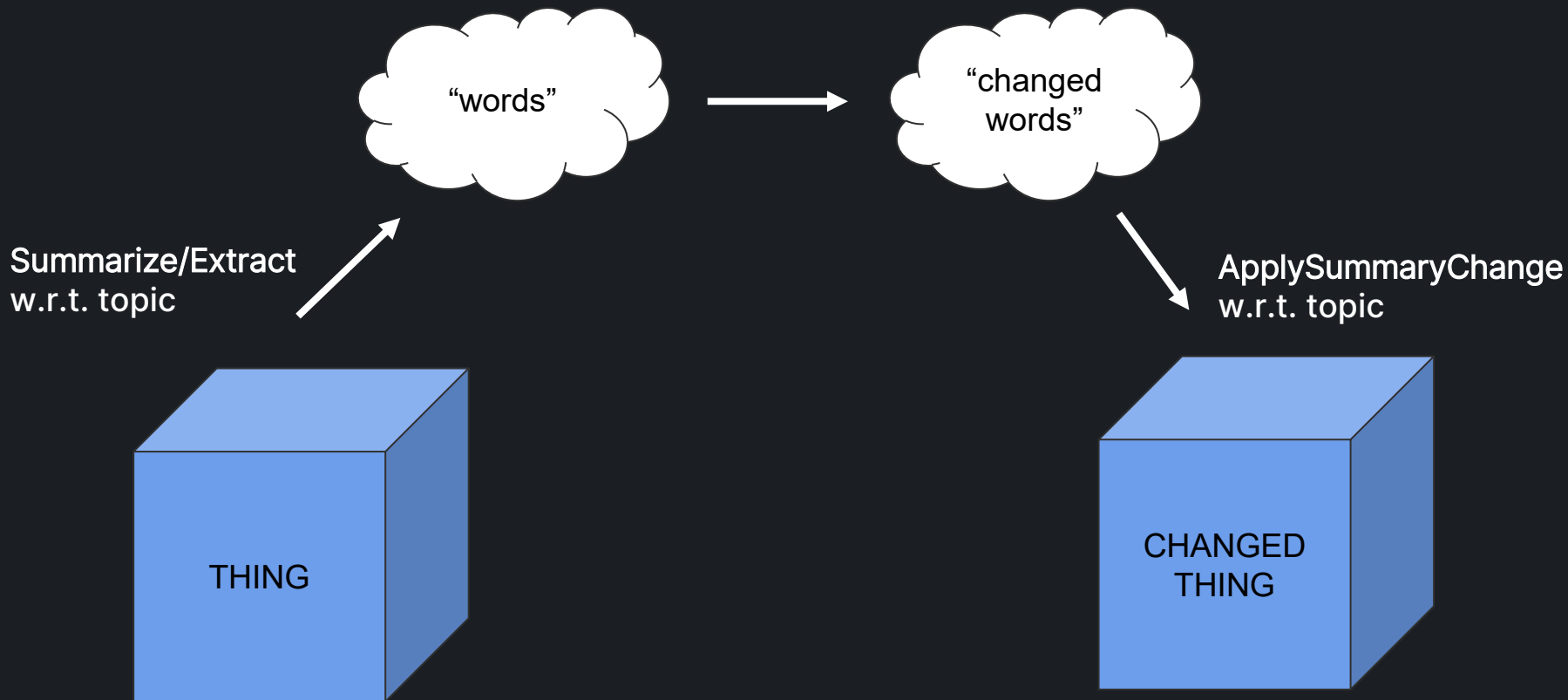
Philosophy



Philosophy



Philosophy



Rationale

- Edit > Create-from-scratch
 - (also, create-from-scratch is a special case of editing)
- People lack the terms of discourse to describe the change they want
 - The pivot pins us down to a particular slice
 - The extraction describes the thing as it is
 - The edit describes how we want it to be



Principles

- Change is described in language
- Specifications are ephemeral
- There are many specifications, one for each topic
- The model both hides and fills in the details



Demo



User Statement

User statement

As a `developer` :

I want to `understand my code and explore changes to achieve some overall change in specification` :

But `I must manually understand and work out the necessary changes` :

This project helps by `allowing me to extract a high-level summary on a topic, edit the summary and apply the change and iterate.`

Unlike `manually searching, understanding and constructing the necessary changes`

Measurable Outcomes (perceived or observed)

- ✓ Users can experience `deeper understanding and quicker prototyping of changes` , so we can directly ask them
- ✓ `deeper understanding and quicker prototyping of changes` is observable, so we can test it with users



Notes

- Summarization the inverse of generation (thanks Eddie!)
 - “Extract” is summarization w.r.t. topic
- “Extract” seems independently useful
 - has similarity with Chat
- “Apply” utilises hallucination well!
 - “Fill in the details, I’ll check what you do”
- Is this a new software methodology/modality?
 - Almost. Sometimes struggle to find a pivot, prefer imperative language - “add this feature”.
 - But this is likely inaccurate, like hacking with a saw not a scalpel.



Challenges

- Deliveries?

- A general form of semantic search & replace in VS Code?
- A new IDE?

- Performance

- “Extract” is summarization and relatively fast
- “Apply” is code-rewriting and slow. UX helps, picking files helps, planning helps. UX

- Performance

- “Extract” is summarization and relatively fast
- “Apply” is code-rewriting and slow. UX helps, picking files helps, planning helps.

